22103

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Illustrate your answers with neat sketches wherever necessary.

(3) Figures to the right indicate full marks.

(4) Assume suitable data, if necessary.

(5) Use of Non-programmable Electronic Pocket Calculator is permissible.

(6) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following : **10**

(a) Simplify $\log_2 14 - \log_2 7$

(b) Using determinant method find the area of triangle whose vertices are (4, 5) (0, 7) & (–1, 1).

(c) Without using calculator find the value of cos (75°).

(d) If the diagonals of a rhombus are 16 cm & 12 cm, find its area.

(e) A cone and cylinder having the same area of the base and having also the same area of curved surfaces. If the height of the cylinder be 2 m, find the slant height of the cone.

(f) The coefficient of variance of a certain distribution is 4 and mean is 125. Find standard deviation.

(g) Find the range and coefficient of range of the following :

80, 40, 120, 90, 50, 200, 180.

2. Attempt any THREE of the following : **12**

(a) Find Inverse of matrix A, by adjoint method

where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

(b) Resolve into partial fraction,

$$\frac{x+3}{(x-1)(x+1)(x+5)}$$

(c) Using Cramer's rule, following equations are obtained as a result of an experiment :

$P_1 + P_2 + P_3 = 6$; $P_2 + 3P_3 = 11$ and $P_1 - 2P_2 + P_3 = 0$.

Find value of P_1, P_2 & P_3.

(d) Calculate standard deviation and coefficient of variance from the following table :

Class Mark	7	8	9	10	11	12	13
Frequency	4	6	9	12	9	6	4

3. Attempt any THREE of the following : **12**

(a) If $\tan\frac{\theta}{2} = \frac{2}{3}$, find value of '$2 \sin\theta + 3 \cos\theta$'.

(b) If $2 \cos 60° \cdot \cos 10° = \cos A + \cos B$

then find A and B.

(c) Simplify $\frac{\cos^2(180° - \theta)}{\sin(-\theta)} + \frac{\cos^2(270° - \theta)}{\sin(180° - \theta)}$.

(d) Prove that $\frac{\sin 3A - \sin A}{\cos 3A + \cos A} = \tan A$.

4. Attempt any THREE of the following : **12**

(a) If $A = \begin{bmatrix} 2 & 4 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

then prove that $A^2 = 3A + 2I$.

(b) Resolve into partial fraction $\frac{3x-2}{(x+2)(x^2+4)}$.

(c) Prove that $8 \sin 20° \cdot \sin 40° \cos 10° = \sqrt{3}$.

(d) Prove that $\frac{\cos 3A - \cos 7A}{\sin 9A + \sin A} = \cos 2A \tan 4A - \sin 2A$.

(e) If $\tan x = \frac{5}{6}$ and $\tan y = \frac{1}{11}$

show that $x + y = 45°$.

5. Attempt any TWO of the following : **12**

(a) Attempt the following :

(i) Find the equation of the line whose x-intercept is 3 and y intercept is 4.

(ii) Find the equation of line passing through the point (2, 3) and having slope $\frac{1}{2}$.

(b) Attempt the following :

(i) Find the equation of the line passing through (2, –3) and parallel to the line $4x - y + 7 = 0$.

(ii) Show that the point (1, 2) is equidistant from $4x - 3y + 7 = 0$ and $5x + 12y = 16$.

(c) Attempt the following :

Find the length of the longest pole that can be placed in a room 12 m long, 9 m broad and 8 m high.

6. Attempt any TWO of the following : **12**

(a) Solve the following equation by using matrix inversion method :

$x + y + z = 6$, $3x - y + 3z = 10$, $5x + 5y - 4z = 3$

(b) Find mean deviation about mean of the following distribution :

C.I.	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
Frequency	4	6	10	18	9	3

(c) Attempt the following :

(i) Find range and co-efficient of range of the following :

C.I.	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79
Frequency	10	15	30	20	15	10

(ii) The data gives mean weight and S.D. for two groups :

Group	Mean Weight	S.D.
A	80 kg	6.0 kg
B	60 kg	7.0 kg

Which of the two groups has greater variability ?
